Vectors

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vector arithmetic in three dimensions**

Find the magnitude of $\mathbf{v} = (6, -3, 2)$, and hence find a unit vector in the same direction.

$|\mathbf{v}| = \sqrt{6^2 + (-3)^2 + 2^2} = \sqrt{49} = 7$. Unit vector: $\frac{1}{7}(6, -3, 2) = \left(\frac{6}{7}, -\frac{3}{7}, \frac{2}{7}\right)$.

2 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 1, -2)$ and $\mathbf{b} = (-1, 4, 2)$, and state whether $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

$\mathbf{a} \cdot \mathbf{b} = 3(-1) + 1(4) + (-2)(2) = -3 + 4 - 4 = -3$. Since $\mathbf{a} \cdot \mathbf{b} \neq 0$, they are not perpendicular.

3 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find the angle between $\mathbf{a} = (1, 2, 2)$ and $\mathbf{b} = (2, -1, 2)$, to the nearest degree.

$\mathbf{a} \cdot \mathbf{b} = 2 - 2 + 4 = 4$. $|\mathbf{a}| = 3$, $|\mathbf{b}| = 3$. $\cos\theta = \frac{4}{9} \approx 0.4444$, so $\theta \approx 64^\circ$.

4 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find the value of k such that $\mathbf{a} = (3, k, -2)$ and $\mathbf{b} = (1, -2, 4)$ are perpendicular.

For perpendicular vectors, $\mathbf{a} \cdot \mathbf{b} = 0$: $3(1) + k(-2) + (-2)(4) = 0 \Rightarrow 3 - 2k - 8 = 0 \Rightarrow -2k = 5 \Rightarrow k = -\frac{5}{2}$.

5 **[Foundational]** • **Target: The vector equation of a line**

Write the vector equation of the line through $(1, 2, -1)$ with direction vector $(2, -1, 3)$.

$\mathbf{r} = (1, 2, -1) + t(2, -1, 3)$, where $t \in \mathbb{R}$.

6 **[Foundational]** • **Target: The vector equation of a line**

Find the vector equation of the line through the points $A(2, 0, 1)$ and $B(5, 3, -2)$.

Direction: $\vec{AB} = (5 - 2, 3 - 0, -2 - 1) = (3, 3, -3)$, or simplified $(1, 1, -1)$. Vector equation: $\mathbf{r} = (2, 0, 1) + t(1, 1, -1)$.

7 **[Foundational]** • **Target: The vector equation of a line**

Find the vector equation of the line through $(0, 1, -2)$ parallel to the line $\mathbf{r} = (3, 0, 1) + t(2, -1, 4)$.

Parallel lines share the same direction vector, so $\mathbf{r} = (0, 1, -2) + t(2, -1, 4)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Find the acute angle between lines with direction vectors $(1, 2, 2)$ and $(2, 1, -2)$, to the nearest degree.

Dot product: $1(2) + 2(1) + 2(-2) = 2 + 2 - 4 = 0$. Since the dot product is zero, the lines are perpendicular: the angle is 90°.

9 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Determine whether the lines $\mathbf{r}_1 = (1, 0, 2) + s(1, 1, 1)$ and $\mathbf{r}_2 = (2, 1, 0) + t(2, 2, 2)$ are parallel, and if so, whether they are the same line.

Direction vectors $(1, 1, 1)$ and $(2, 2, 2) = 2(1, 1, 1)$ are scalar multiples, so the lines are parallel. Testing whether $(2, 1, 0)$ lies on line 1: $(1 + s, s, 2 + s) = (2, 1, 0) \Rightarrow s = 1$ from the y-coordinate, but then the x-coordinate gives $1 + 1 = 2$ ✓ while the z-coordinate gives $2 + 1 = 3 \neq 0$ ✗. So the point does not lie on line 1 — the lines are parallel but distinct, and never meet.

10 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Show that the points $A(1, 2, 3)$, $B(3, 6, 7)$, and $C(5, 10, 11)$ are collinear.

$\vec{AB} = (3 - 1, 6 - 2, 7 - 3) = (2, 4, 4)$. $\vec{AC} = (5 - 1, 10 - 2, 11 - 3) = (4, 8, 8) = 2\vec{AB}$. Since $\vec{AC}$ is a scalar multiple of $\vec{AB}$ and both start at A, the three points lie on the same line, so A, B, C are collinear.

11 [Intermediate] • Target: Intersecting and skew lines

Determine whether the lines $\mathbf{r}_1 = (2, 1, 3) + s(1, 0, -1)$ and $\mathbf{r}_2 = (0, 2, 5) + t(1, 1, -2)$ intersect.

Equate components: $2 + s = t$ (x); $1 = 2 + t \Rightarrow t = -1$ (y); then $s = t - 2 = -3$. Check the z-coordinate: line 1 gives $3 - s = 3 - (-3) = 6$; line 2 gives $5 - 2t = 5 - 2(-1) = 7$. Since $6 \neq 7$, the equations are inconsistent, and since the direction vectors $(1, 0, -1)$ and $(1, 1, -2)$ are not scalar multiples (not parallel), the lines are skew.

12 [Intermediate] • Target: Intersecting and skew lines

Find the point of intersection of the lines $\mathbf{r}_1 = (2, 1, 1) + s(1, 1, 2)$ and $\mathbf{r}_2 = (0, 5, 3) + t(2, -1, 1)$.

Equate components: $2 + s = 2t$ (x); $1 + s = 5 - t$ (y); $1 + 2s = 3 + t$ (z). From the y-equation, $s = 4 - t$. Substituting into the x-equation: $2 + (4 - t) = 2t \Rightarrow 6 - t = 2t \Rightarrow t = 2$, so $s = 2$. Checking the z-equation: $1 + 2(2) = 5$ and $3 + 2 = 5$ ✓. Point of intersection (using $s = 2$ in line 1):
$(2 + 2, 1 + 2, 1 + 4) = (4, 3, 5)$.

13 [Intermediate] • Target: The equation of a plane

Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.

The plane has the form $2x - y + 3z = d$. Substituting the point: $2(1) - 2 + 3(-1) = 2 - 2 - 3 = -3$. Equation: $2x - y + 3z = -3$.

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14 [Intermediate] • Target: The equation of a plane

Find the Cartesian equation of the plane containing the points $A(2, 0, 0)$, $B(0, 3, 0)$, and $C(0, 0, 4)$.

$\vec{AB} = (-2, 3, 0)$, $\vec{AC} = (-2, 0, 4)$. Normal $= \vec{AB} \times \vec{AC} = (3 \cdot 4 - 0 \cdot 0,\ 0 \cdot (-2) - (-2) \cdot 4,\ (-2) \cdot 0 - 3 \cdot (-2)) = (12, 8, 6)$, simplifying to $(6, 4, 3)$. Plane: $6x + 4y + 3z = d$; substituting A: $6(2) = 12 = d$. Equation: $6x + 4y + 3z = 12$ (matching the intercept form $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The equation of a plane • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Find the Cartesian equation of the plane through $(1, -1, 2)$ perpendicular to the vector $(3, 0, -2)$.

The plane has the form $3x - 2z = d$. Substituting the point: $3(1) - 2(2) = 3 - 4 = -1$. Equation: $3x - 2z = -1$.

16 [Challenging] • Target: The equation of a plane • Cross-Lesson: 9.01 Complex Numbers

Find the Cartesian equation of the plane through the points $A(2, 1, 0)$, $B(0, 2, 1)$, and $C(1, 0, 3)$.

$\vec{AB} = (-2, 1, 1)$, $\vec{AC} = (-1, -1, 3)$. Normal $= \vec{AB} \times \vec{AC} = (1 \cdot 3 - 1 \cdot (-1),\ 1 \cdot (-1) - (-2) \cdot 3,\ (-2) \cdot (-1) - 1 \cdot (-1)) = (4, 5, 3)$. Plane: $4x + 5y + 3z = d$; substituting A: $4(2) + 5(1) = 13 = d$. Equation: $4x + 5y + 3z = 13$.

17 [Challenging] • Target: Angles between lines, planes, and pairs of planes • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Find the angle between the line with direction $(2, 1, 2)$ and the plane $2x - y + 2z = 6$, to the nearest degree.

Plane normal: $\mathbf{n} = (2, -1, 2)$. $\mathbf{d} \cdot \mathbf{n} = 2(2) + 1(-1) + 2(2) = 4 - 1 + 4 = 7$. $|\mathbf{d}| = 3$, $|\mathbf{n}| = 3$. $\sin\theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|} = \frac{7}{9} \approx 0.778$, so $\theta \approx 51^\circ$.

18 [Challenging] • Target: Angles between lines, planes, and pairs of planes • Cross-Lesson: 10.01 Numerical Methods

Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.

Normals: $\mathbf{n}_1 = (1, 2, -1)$, $\mathbf{n}_2 = (2, -1, 1)$. $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 2 - 1 = -1$. $|\mathbf{n}_1| = \sqrt{6}$, $|\mathbf{n}_2| = \sqrt{6}$. $\cos\theta = \frac{|-1|}{6} \approx 0.167$, so $\theta \approx 80^\circ$.

19 [Challenging] • Target: Distances and same-side reasoning • Cross-Lesson: 11.01 Differential Equations

Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using $d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}$.

$d = \frac{|2(3) - 1(1) + 2(-2) - 6|}{\sqrt{4 + 1 + 4}} = \frac{|6 - 1 - 4 - 6|}{3} = \frac{5}{3}$.

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20 [Challenging] • Target: Distances and same-side reasoning • Cross-Lesson: 12.01 Poisson Distribution

Find the foot of the perpendicular from $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$, and the shortest distance from P to the line.

A general point on the line is $(1 + t,\ t,\ t)$. The vector from this point to P is $(3 - t,\ -t,\ 1 - t)$, which must be perpendicular to $(1, 1, 1)$: $(3 - t) - t + (1 - t) = 0 \Rightarrow 4 - 3t = 0 \Rightarrow t = \frac{4}{3}$. Foot of perpendicular: $\left(\frac{7}{3}, \frac{4}{3}, \frac{4}{3}\right)$. Vector from P to this foot: $\left(-\frac{5}{3}, \frac{4}{3}, \frac{1}{3}\right)$, with magnitude $\sqrt{\frac{25 + 16 + 1}{9}} = \sqrt{\frac{42}{9}} = \frac{\sqrt{42}}{3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vector arithmetic in three dimensions

A rectangular garden is to be enclosed by a fence. One side of the garden is adjacent to a long straight river, and requires no fencing. The other three sides require fencing, and the total length of the fence available is 120 meters. Let the side of the garden parallel to the river have length x meters, and let the two sides perpendicular to the river each have length y meters. The area of the garden is given by $A(x, y) = xy$. \\ Find the dimensions x and y that maximize the area of the garden.

a. **x = 60 m, y = 30 m** ✓

b. x = 30 m, y = 60 m

c. x = 40 m, y = 40 m

d. x = 80 m, y = 20 m

The total length of the fence is 120 meters. The fence is used for the three sides of the rectangle not along the river. Thus, the constraint equation is $x + 2y = 120$. \\ We want to maximize the area $A = xy$. From the constraint equation, we can express x in terms of y: $x = 120 - 2y$. \\ Substitute this expression for x into the area formula: $A(y) = (120 - 2y)y = 120y - 2y^2$. \\ To find the maximum area, we need to find the critical points of $A(y)$ by taking the derivative with respect to y and setti.

22 [Foundational] • Target: The scalar (dot) product and the angle between vectors

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

a. $\langle 3, 4 \rangle$ ✓

b. $\langle -3, -4 \rangle$

c. $\langle 7, 10 \rangle$

d. $\langle 1, 4 \rangle$

To find the component form of a vector $\vec{PQ}$ from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$, we subtract the coordinates of the initial point P from the coordinates of the terminal point Q. The formula is $\vec{PQ} = \langle x_2 - x_1, y_2 - y_1 \rangle$. Given $P = (2, 3)$ and $Q = (5, 7)$, we have $x_1 = 2$, $y_1 = 3$, $x_2 = 5$, and $y_2 = 7$. Applying the formula: $\vec{PQ} = \langle 5 - 2, 7 - 3 \rangle$ $\vec{PQ} = \langle 3, 4 \rangle$ Therefore, the component form of the vector $\vec{PQ}$ is.

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23 [Foundational] • Target: The vector equation of a line

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:

a. $|\mathbf{b} - \mathbf{a}|$

b. $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$ ✓

c. $|\mathbf{d}_1 - \mathbf{d}_2|$

d. 0

Applying Vectors and Vector-Valued Functions principles, Skew line distance formula. Therefore, the verified correct option is $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$.

24 [Foundational] • Target: Classifying lines: parallel, perpendicular and collinear points

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 20 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

a. 18

b. **20** ✓

c. 25

d. 22

By symmetry (each pair at 120°), the third force balances the resultant of the two equal forces: $F_3 = F_1 = 20$ N (standard 120°-120°-120° symmetric equilibrium).

25 [Foundational] • Target: Intersecting and skew lines

A line has direction $\mathbf{d} = (2;\ -1; 0)$ and a plane has normal $\mathbf{n} = (1; 2; 2)$. Find the angle between the line and the plane.

a. 30°

b. 45°

c. 0° ✓

d. 90°

Working through this vectors step: $\mathbf{d} \cdot \mathbf{n} = 0 \Rightarrow \mathbf{d}$ lies in the plane $\Rightarrow$ angle $= 0^\circ$

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26 [Foundational] • Target: The equation of a plane

Dot product $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\theta$. For perpendicular vectors, $\mathbf{a} \cdot \mathbf{b} =$

a. 1

b. -1

c. 0 ✓

d. $|a||b|$

Applying Vectors and Vector-Valued Functions principles, $\cos 90^\circ = 0$. Therefore, the verified correct option is 0.

27 [Foundational] • Target: Angles between lines, planes, and pairs of planes

Find the angle between planes $x + y = 0$ and $x + z = 0$:

a. $\pi/3$ ✓

b. 0

c. $\pi/2$

d. $\pi/4$

Normals $(1, 1, 0)$ and $(1, 0, 1)$; $\cos\theta = 1/2 \Rightarrow \theta = \pi/3$.

28 [Foundational] • Target: Distances and same-side reasoning

Line of intersection of two non-parallel planes: direction vector $=$

a. sum of normals

b. **cross product of normals** ✓

c. difference of normals

d. dot product of normals

Intersection direction $= \mathbf{n}_1 \times \mathbf{n}_2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a drone's flight path

Two vectors are anti-parallel if angle between them is:

a. 0

b. $\pi/4$

c. π ✓

d. $\pi/2$

Anti-parallel: $\theta = \pi$ (opposite directions).

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30 [Intermediate] • Target: Vector arithmetic in three dimensions

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$ ✓

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

The position vectors of particle A and particle B at time t are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$. The vector representing the displacement from particle B to particle A at time t is $\mathbf{d}(t) = \mathbf{r}_A(t) - \mathbf{r}_B(t) = (2t - (3 + 5t); (4t - 1) - (1 - 2t)) = (-3t - 3; 6t - 2)$. The distance between the particles at time t is the magnitude of this displacement vector.

31 [Intermediate] • Target: The scalar (dot) product and the angle between vectors

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

a. (-3, 2)

b. (-2, -3)

c. **(2, -3)** ✓

d. (-3, -2)

The original points are A(1, 2), B(4, 2), and C(2, 5). The curve $y = f(x)$ is a parabola. Since A and B have the same y-coordinate, the axis of symmetry is $x = \frac{1+4}{2} = 2.5$. The equation of the parabola can be written as $y = a(x - 2.5)^2 + k$. Substituting C(2, 5): $5 = a(2 - 2.5)^2 + k \Rightarrow 5 = 0.25a + k$. Step 1: Reflection in the line $y = x$. This transformation swaps the x and y coordinates. A'(2, 1), B'(2, 4), C'(5, 2). Step 2: Translation by the vector $(-3; 0)$. This subtracts 3.

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32 [Intermediate] • Target: The vector equation of a line

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.

a. **-6** ✓

b. 6

c. 3

d. -3

Two vectors are parallel if one is a scalar multiple of the other. Therefore, there must exist a scalar c such that $\mathbf{u} = c\mathbf{v}$. This gives us the equations: $3 = c(-2)$ $k = c(4)$ From the first equation, we can solve for c: $c = \frac{3}{-2} = -\frac{3}{2}$ Now substitute this value of c into the second equation to find k: $k = \left(-\frac{3}{2}\right)(4)$ $k = -\frac{12}{2}$ $k = -6$ Therefore, the value of k is -6. A common algebraic trap is to mistakenly set the components equal with.

33 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Consider the points A(2, 3) and B(8, 7) in the Cartesian plane. The vector $\vec{AB}$ can be represented as $\mathbf{v}$. Find the components of $\mathbf{v}$.

a. **(6, 4)** ✓

b. (10, 10)

c. (-6, -4)

d. (4, 6)

To find the vector $\vec{AB}$, we subtract the coordinates of the initial point A from the coordinates of the terminal point B. Let A = (x■, y■) = (2, 3) and B = (x■, y■) = (8, 7). The vector $\vec{AB}$ is given by $\mathbf{v} = (x_2 - x_1, y_2 - y_1)$. Substituting the given coordinates: $\mathbf{v} = (8 - 2, 7 - 3) = (6, 4)$. Therefore, the components of the vector $\vec{AB}$ are (6, 4). Common errors include subtracting in the reverse order (which would give (-6, -4)) or adding the coordinates (which would give (10.

34 [Intermediate] • Target: Intersecting and skew lines

Which is an equation of the plane through $(-3,\ 1,\ 4)$, $(-2,\ 2,\ 4)$ and $(1,\ -1,\ 1)$?

a. $x - y + 2z = -4$

b. $x + y + 2z = 6$

c. $x - y + 2z = 4$ ✓

d. $x + y = -2$

A normal vector is $\vec{PQ} \times \vec{PR}$, a multiple of $\langle 1,\ -1, 2\rangle$. Using the point $(-3,\ 1,\ 4)$ gives $x - y + 2z = 4$.

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35 **[Intermediate]** • **Target: The equation of a plane**

Line integral of scalar field: $\int_C f\,ds = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|dt$. For $f = 1$, $\int_C 1\,ds =$

a. area

b. volume

c. 0

d. **arc length ✓**

Applying Vectors and Vector-Valued Functions principles, $\int_C ds = L$ (arc length). Therefore, the verified correct option is arc length.

36 **[Intermediate]** • **Target: Angles between lines, planes, and pairs of planes**

Where does the line $\mathbf{r}(t) = \langle 3, -3, -1\rangle + t\langle 0, 0, 1\rangle$ meet the plane $3x + y + 3z = 4$?

a. $\left(3, -3, \frac{1}{3}\right)$

b. $\left(3, -3, -\frac{2}{3}\right)$ ✓

c. $\left(3, -3, -\frac{4}{3}\right)$

d. $(3, -3, -1)$

Substitute the line into the plane: $3t + 3 = 4$, so $t = \frac{1}{3}$ and the point is $\left(3, -3, -\frac{2}{3}\right)$.

37 **[Intermediate]** • **Target: Distances and same-side reasoning**

A line has vector equation $\mathbf{r} = (-1; 4; 6) + t(-1; 4; 2)$. Find the point when $t = -4$.

a. $(3, -12, -2)$ ✓

b. $(4, -10, -3)$

c. $(2, -10, 0)$

d. $(5, -14, 0)$

Working through this vectors step: $\mathbf{r} = (-1; 4; 6) + (-4)(-1; 4; 2) = (3, -12, -2)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A well-designed word problem: a drone's flight path • Cross-Lesson: 10.01 Numerical Methods

Find the angle between gradient of $f = x^2 + y^2 + z^2$ at $(1, 1, 1)$ and gradient of $g = x + 2y + 3z$:

a. 0

b. $\theta = \arccos \frac{6}{\sqrt{12}\sqrt{14}}$ ✓

c. $\pi/2$

d. π

$ablaf = (2, 2, 2)$, $ablag = (1, 2, 3)$; $\cos\theta = (2 + 4 + 6)/(\sqrt{12}\sqrt{14}) = 12/(\sqrt{168})$. Choose A.

39 [Challenging] • Target: Vector arithmetic in three dimensions • Cross-Lesson: 11.01 Differential Equations

Line 1: $\mathbf{r} = (-2;\ -4; 2) + s(-2; 3;\ -1)$. Line 2: $\mathbf{r} = (0;\ -1; 4) + t(-4; 6;\ -2)$. Are the lines parallel?

a. Yes ✓

b. No

c. They are the same line

d. They intersect at one point

Direction vector 2 = 2 × direction vector 1, so the lines are parallel.

40 [Challenging] • Target: The scalar (dot) product and the angle between vectors • Cross-Lesson: 12.01 Poisson Distribution

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-5;\ -2; 2)$ passing through $P(-3,\ -4,\ -3)$.

a. $-5x - 2y + 2z = 20$

b. $-5x - 2y + 2z = 18$

c. $-5x - 2y + 2z = 17$ ✓

d. $-5x - 2y + 2z = 14$

Working through this vectors step: $d = \mathbf{n} \cdot P = (-5)(-3) + (-2)(-4) + (2)(-3) = 17$

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41 [Challenging] • Target: The vector equation of a line • Cross-Lesson: 13.01 Continuous Random Variables

Equation of plane through $(1, 0, 0)$ with normal $\mathbf{n} = (1, 2, 3)$:

a. $x + 2y + 3z = 1$ ✓

b. $x + 2y + 3z = 0$

c. $2x + y + 3z = 1$

d. $x = 1$

$\mathbf{n} \cdot (\mathbf{r} - P_0) = 0 \Rightarrow (x - 1) + 2y + 3z = 0 \Rightarrow x + 2y + 3z = 1.$

42 [Challenging] • Target: Classifying lines: parallel, perpendicular and collinear points • Cross-Lesson: 14.01 Sampling & Estimation

Find the angle between lines with direction vectors $\mathbf{d_1} = (1; 4; 8)$ and $\mathbf{d_2} = (4; -1; 0)$.

a. 90° ✓

b. 45°

c. 60°

d. 30°

$\mathbf{d_1} \cdot \mathbf{d_2} = 0 = 0 \Rightarrow \theta = 90^\circ$

43 [Challenging] • Target: Intersecting and skew lines • Cross-Lesson: 15.01 Hypothesis Testing

Two planes are parallel if their normal vectors are:

a. **parallel (scalar multiples)** ✓

b. perpendicular

c. equal in magnitude

d. zero

Parallel planes: normals are scalar multiples.

44 [Challenging] • Target: The equation of a plane • Cross-Lesson: 2.01 Exponentials & Logarithms

Two vectors are given by $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$. If $\vec{u}$ and $\vec{v}$ are orthogonal, find the value of the scalar parameter k.

a. 3/4

b. **4/3** ✓

c. -3/4

d. -4/3

Two vectors $\vec{u}$ and $\vec{v}$ are orthogonal if and only if their dot product is zero. The dot product of $\vec{u} = u_1\hat{i} + u_2\hat{j}$ and $\vec{v} = v_1\hat{i} + v_2\hat{j}$ is given by $\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2$. Given $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$, their dot product is: $\vec{u} \cdot \vec{v} = (3)(-1) + (k)(4)$ $\vec{u} \cdot \vec{v} = -3 + 4k$ For the vectors to be orthogonal, their dot product must be zero: $-3 + 4k = 0$ $4k = 3$.

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45 [Challenging] • Target: Angles between lines, planes, and pairs of planes • Cross-Lesson: 3.01 Advanced Trigonometry

Consider the points $A(1, 2, 3)$ and $B(4, 6, 9)$. Find the vector $\vec{AB}$ in component form.

a. $(3; 4; 6)$ ✓

b. $(5; 8; 12)$

c. $(-3; -4; -6)$

d. $(4; 6; 9)$

To find the vector $\vec{AB}$, we subtract the coordinates of the initial point A from the coordinates of the terminal point B. Let $A = (x_1, y_1, z_1) = (1, 2, 3)$ and $B = (x_2, y_2, z_2) = (4, 6, 9)$. The vector $\vec{AB}$ is given by $(x_2 - x_1; y_2 - y_1; z_2 - z_1)$. Substituting the given coordinates: $\vec{AB} = (4 - 1; 6 - 2; 9 - 3) = (3; 4; 6)$. Therefore, the vector $\vec{AB}$ in component form is $(3; 4; 6)$.

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